

no PyTorch or scikit-learn past this line

OFFLINE · batch

notebooks/01_eda.ipynb

Steps 1-13: grid searches, ablations,
every figure and number in the report



selected hyperparameters

scripts/export_recommendations.py

refits the Nested Hybrid on train+val,
scores all 610 users (needs torch)



data/recommendations_export.json

top-20 per user · each user's own ratings
Most-Popular cold-start fallback list



ONLINE · request path

api/main.py (FastAPI)

RecommendationService loads the JSON once;
a lookup is a dict hit



web/index.html

user picker, held-out-hit badges,
fallback banner



Unknown user (e.g. userId 611)

never scored by the batch job, so there is nothing
to look up. Returns 200 with `is_fallback: true` and
the Most-Popular list - a baseline whose quality is
already measured (test NDCG@10 0.1549), not a 404.